

Topic:-

Study osmosis using parchment paper or a similar membrane.

Introduction:-

Every morning, when dry raisins (kismis) are soaked in plain water, they swell up plump and full of juice by evening. Conversely, when fresh cucumber slices are sprinkled with salt, water droplets immediately weep out of them. Have you ever wondered what physical force drives this movement of water? It occurs because of a vital biological process known as **Osmosis**.

In Chapter 10 of our Class 8 Science curriculum, we learn that living cell membranes are selectively permeable (semi-permeable). A semi-permeable membrane contains microscopic pores that permit small water molecules to pass freely, while blocking larger dissolved solute molecules like sugar and salt. Osmosis is the spontaneous passage of solvent (water) molecules from a region of higher water concentration (dilute solution) to a region of lower water concentration (concentrated solution) through a semi-permeable membrane. In this project, we demonstrated osmosis using parchment paper and sugar solution.

❖ Three Classes of Solutions in Osmosis:-

- **Hypotonic:** Liquid outside has more water than inside cell; water enters and cell swells.
- **Hypertonic:** Liquid outside has less water than inside cell; water exits and cell shrinks.
- **Isotonic:** Water concentration is identical on both sides; zero net water movement.

Aim & Laboratory Apparatus:-

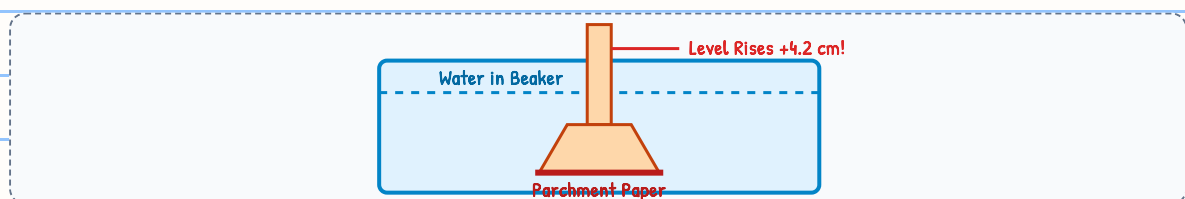
Aim: To set up a thistle funnel osmometer using parchment paper as a semi-permeable membrane, and measure the rate of water intake into a concentrated sugar solution.

❖ *Objectives of the Experiment:-*

- To construct a leak-proof semi-permeable membrane barrier using parchment paper.
- To record the gradual rise of liquid level in the funnel stem at 30-minute intervals.
- To explain how plant root hairs absorb soil water via natural osmosis.

❖ *Apparatus Required:-*

- Glass thistle funnel (or a cut inverted plastic bottle with narrow neck), laboratory stand with clamp.
- Sheet of genuine parchment paper (or cellophane sheet), strong cotton thread, rubber band.
- Glass beaker (500 mL), distilled water, concentrated sugar syrup, red food dye, ruler, and marker pen.



Step-by-Step Experimental Procedure:-

We executed the laboratory osmosis experiment through these rigorous steps:

- **Step 1: Preparing Concentrated Solution:** Dissolved 4 tablespoons of commercial sugar in 100 mL of warm water and stirred in 2 drops of red food dye for clear visibility.
- **Step 2: Securing the Membrane:** Soaked a circular sheet of parchment paper in water for 2 minutes to make it supple. Stretched it tightly over the wide mouth of the thistle funnel and tied it firmly with waxed thread and a tight rubber band.
- **Step 3: Checking for Leaks:** Inverted the funnel and inspected for water droplets. Ensured zero leaks at the parchment rim.
- **Step 4: Filling the Funnel:** Carefully poured the red sugar solution through the stem until it completely filled the bulb and rose 2.0 cm up the stem.
- **Step 5: Mounting in Water:** Clamped the funnel upright in a beaker filled with plain water, ensuring the parchment mouth was submerged. Marked the starting level with a waterproof marker.

Observations & Timed Measurement Table:-

We monitored and recorded the liquid column height every 30 minutes over a 2-hour duration:

<i>Elapsed Time</i>	<i>Column Height in Stem</i>	<i>Net Rise Observed</i>	<i>Physical Laboratory Observation</i>
0 Min (Start)	2.0 cm	0.0 cm	Initial level marked; solution calm
30 Minutes	3.2 cm	+ 1.2 cm	Noticeable upward creep; red dye dilute
60 Minutes (1 Hr)	4.5 cm	+ 2.5 cm	Continuous steady rise of colored water
90 Minutes	5.6 cm	+ 3.6 cm	Rise slows slightly as hydrostatic pressure mounts
120 Minutes (2 Hr)	6.2 cm	+ 4.2 cm	Level reaches steady equilibrium state

Scientific Analysis & Discussion:-

Analyzing our empirical data verifies the thermodynamics of osmosis:

- **Why Did the Water Level Rise?:** Pure water inside the beaker has a higher chemical water potential (100% water concentration). The sugar solution inside the funnel has lower water potential because sugar molecules occupy space. Therefore, water molecules naturally diffused across the parchment paper into the funnel.
- **Why Did Sugar Not Diffuse Out?:** Parchment paper has microscopic cellulose pore diameters smaller than 2 nanometers. Tiny water molecules (diameter ~ 0.28 nm) slip through easily, while bulky sucrose sugar molecules (diameter ~ 1.0 nm) are trapped inside!
- **Concept of Osmotic Pressure:** As the water column rises to 6.2 cm, the downward weight of the liquid exerts hydrostatic pressure. When this downward pressure matches the inward osmotic drive, net water movement stops, reaching dynamic equilibrium.

❖ *Control Experiment Validation:-*

In our control setup using plain water inside the funnel, the liquid level showed 0.0 cm change over 2 hours, proving that sugar concentration drives the suction force!

Everyday Applications & Biological Importance:-

Osmosis is not just a laboratory demonstration; it powers life processes across our planet:

- **Plant Root Absorption:** Billions of root hairs absorb groundwater from soil via endosmosis and pump it upward to tall tree leaves without mechanical motors.
- **Food Preservation in Jams & Pickles:** Adding heavy sugar to fruit jams or heavy salt to raw mango pickles creates a hypertonic environment. When bacteria touch the food, water is sucked out of their cells by exosmosis, dehydrating and killing the germs!
- **Medical Saline Drips (0.9% NaCl):** Hospital IV drips must be strictly isotonic with human blood. If pure water were injected, red blood cells would absorb water, swell, and burst dangerously (hemolysis).

❖ Keeping Wilted Vegetables Fresh:-

Sprinkling cold tap water over wilted spinach leaves restores crispness within 15 minutes as water re-enters plant cells via endosmosis!

What We Learned & Conclusion:-

- Osmosis is the net movement of water from higher water potential to lower water potential across a semi-permeable membrane.
- Parchment paper acts as an effective semi-permeable membrane that lets water pass while blocking sugar solutes.
- The liquid column rose by 4.2 cm in 2 hours, demonstrating significant osmotic pressure.
- Osmosis governs water uptake in plant roots, human cellular hydration, and traditional food preservation methods.

Conclusion:-

Our thistle funnel experiment successfully demonstrated the phenomenon of osmosis using parchment paper. Watching the colored sugar solution rise steadily by 4.2 cm provided clear physical evidence of osmotic force. This foundational experiment helped us understand how plants drink groundwater and how our own cells regulate vital fluid balance every second of our lives.

References:-

- Maharashtra State Board Class 8 General Science Textbook (Chapter 10: Cell and Cell Organelles).
- Standard 8 Science Laboratory Manual – Plant Physiology and Osmosis.
- NCERT Class 8 Science Guide: Cellular Transport and Membranes.